

CRISPR-Cas9: A Testament to the Value of Basic Research

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Abstract

The CRISPR-Cas9 system is a programmable method of editing DNA sequences and its discovery has transformed the fields of medicine, agriculture, and biotechnology, to name a few. While it has revolutionized modern gene editing approaches, its discovery was not the result of targeted studies. Rather, it was based on years of basic scientific research and the motivation and curiosity of passionate investigators to understand basic biological phenomena. By highlighting key studies, this review traces the historical and experimental trajectory from prior gene-editing techniques, such as restriction enzymes and zinc finger nucleases (ZFNs), to the discovery of the CRISPR-Cas9 system. The story of CRISPR-Cas9, from being investigated as a form of bacterial adaptive immunity to the dawning realization of its massive potential, symbolizes the value and necessity of continued investment in basic scientific research.

Introduction

Since his inauguration on January 20th, President Trump and his administration have enacted policies that significantly reduced federal support for science, signing executive orders that froze billions of dollars in research funding (Glenza, 2025; Nature, 2025). These actions have called into question the national commitment to the advancement of scientific knowledge, once deemed an “essential key to our security as a nation” (Tollefson et al., 2025). In a time when the importance of science has somehow become debatable, basic research, or “a systematic study directed toward greater knowledge of phenomena without specific applications in mind” is under even greater threat, as parties interested in shaving off government inefficiencies may view basic research studies as aimless ventures that waste taxpayer dollars (NIAID, 2023). However, as evidenced by our understanding of the revolutionary gene-editing CRISPR-Cas9 system, which can be programmed with RNA to make precise, sequence-specific cuts in DNA, this view of basic research could not be further from the truth. The discovery of CRISPR-Cas9 has opened worlds of new possibilities in fields such as medicine, agriculture, and

bioenergy, and it was discovered through basic research.

Prior Approaches to Gene Editing

A gene-editing tool that could be programmed to modify specific sequences of DNA was sought after by biologists for many decades. At the heart of potential gene-editing technologies were nucleases, enzymes able to cleave DNA or RNA, which are found in bacterial restriction enzymes. A vast range of restriction enzymes exist in nature, each of which recognizes and cleaves its own unique DNA sequence, but they cannot be modified to recognize different sequences. Some researchers attempted to create a programmable nuclease by linking zinc finger proteins, a class of DNA-binding proteins that recognize specific nucleotides, to the Fok I endonuclease, creating a zinc finger-nuclease (ZFN) hybrid (Kim et al., 1996; Urnov et al., 2005). The Kim group confirmed with SDS/PAGE that cleavage at the desired sequence was achieved with their “hybrid restriction enzymes” (Kim et al.). However, in one of their sample lanes demonstrating the success of the CP-QDR ZFN, there were two additional products present (at ~9.2 kb & ~15 kb) that were not addressed or acknowledged in the article. This calls into question the specificity and fidelity of the CP-QDR ZFN. The Urnov group advanced the applications of ZFNs by demonstrating that these enzymes could be used to create double-stranded breaks in human lymphoblast cells. These types of breaks induce homologous recombination (HR), a natural DNA repair process that uses a matching DNA template to fix broken strands, to correct mutations in the gene responsible for the SCID genetic disorder (IL2R γ). Successful HR introduced a restriction enzyme cut site into the genomic DNA, which after being digested, presented itself as a distinct band on PAGE: using this method, they found that HR occurred at a frequency of 18% (Urnov et al.). Using RT-PCR, a method used to detect RNA expression levels, they confirmed that the production of γ C mRNA was restored in cells where HR was able to successfully restore both copies of the IL2R γ gene, demonstrating the therapeutic potential of ZFNs (Urnov et al.). However, a few inherent problems with

ZFNs remained. First, to target a novel sequence of DNA, a new ZFN must be made from scratch by linking a set of zinc finger proteins to Fok I, which is a laborious and time consuming process. Next, the specificity and efficiency of ZFN cleavage is not robust, as demonstrated by the additional CP-QDR products and the low HR frequency.

Discovery of CRISPR Arrays

Unlike ZFNs, which require labor-intensive protein engineering for each new target, the CRISPR-Cas9 system can be easily reprogrammed by simply altering the guiding RNA sequence, allowing for faster and more versatile gene editing. Clustered, regularly interspaced, short palindromic repeats (CRISPR) are short DNA repeat sequences separated by spacer DNA sequences; together, these stretches of CRISPR and spacer sequences are known as CRISPR arrays. To elucidate the function of these motifs, Francisco Mojica, a microbiologist at the University of Alicante in Spain, used BLAST, a bioinformatics technique, to identify any homology that these spacer sequences shared with known genes and found significant similarities between 47 spacers and bacteriophage genomes (Mojica et al., 2005, Table 2). This finding, along with previous studies showing that strains of phage-resistant bacteria containing a spacer similar to a viral gene were resistant to said virus, led Mojica to be the first to suggest a relationship between the CRISPR array and bacterial adaptive immunity. Mojica's findings and hypotheses were read by food science biologists working at Danisco, one of the world's largest producers of food ingredients and pharmaceutical excipients, who observed that some of the bacteria that they were using as dairy cultures were randomly developing resistance to phage attacks. Could it be possible that, in accordance with Mojica's hypothesis, the CRISPR array was somehow involved in conferring resistance to these mutant strains? After sequencing the CRISPR loci of the phage-resistant strains, the Danisco researchers found that the resistant strains contained spacers homologous to genes found within the bacteriophages responsible for the phage attacks on their dairy cultures (Barrangou et al., 2007). To determine whether this exciting association between spacer and phage resistance was due to a causal relationship, they altered the CRISPR locus both by adding homologous spacers to a non-resistant strain, which granted resistance, and deleting the spacers from a resistant strain, which restored susceptibility to phage attack. This established that the spacers were responsible for phage resistance. While their primary focus was on the spacer DNA sequence, they also generated another mutant in which they disrupted the *cas5* gene, which encodes an enzyme containing an HNH-type nuclease motif. This strain lost its resistance, despite

containing the spacers responsible for phage resistance. They didn't know it at the time, but the Danisco researchers had identified the nuclease responsible for CRISPR activity.

CRISPR-Cas as a Programmable System

In addition to functioning as a method of adaptive immunity, CRISPR was also demonstrated to interfere with horizontal gene transfer (HGT) between bacteria (Marraffini & Sontheimer, 2008). After observing a CRISPR locus in a clinically isolated strain of *Staphylococcus aureus*, the researchers were interested in the requirements for HGT interference, specifically the target of CRISPR interference. After introducing silent mutations to the target gene in the conjugative plasmid, which was identified by its homology with a spacer in the isolated strain's CRISPR locus, CRISPR interference activity was lost. To determine whether this was the result of DNA or mRNA interference, a self-splicing intron was introduced into the target gene, modifying the DNA target, but leaving the mRNA product unchanged. This change disrupted CRISPR interference, indicating that the target of CRISPR is DNA. The findings about the DNA target of CRISPR and the importance of the *cas5* HNH nuclease-like enzyme for phage resistance came together when Garneau et al. were able to demonstrate that the CRISPR-Cas system was cleaving DNA (2010). After introducing a plasmid containing proto-spacers (pNT1) to *S. thermophilus*, which contains its own CRISPR locus, they were able to use both gel electrophoresis and Sanger sequencing to confirm that the pNT1 plasmid was being precisely cut three bases upstream of the proto-spacer adjacent motif (a PAM sequence, essential for CRISPR-Cas9 target recognition). When *cas5* was disrupted, in accordance with Barrangou's findings, cleavage of pNT1 failed. Garneau's group concluded their paper by suggesting that the CRISPR-Cas system can be used to generate organisms with increased bacteriophage resistance, not knowing that their results would soon help redefine the boundaries of gene editing.

Conclusion

The programmability of the CRISPR-Cas system was discovered when the specificity of the *cas5* enzyme (now known as *cas9*) was determined to be dictated by crRNA, the transcribed products of spacer sequences (Jinek et al., 2012). RNA constructs ('chimeras') were synthesized, linking crRNA to tracrRNA, which activates crRNA, and when they were used to program the Cas9 nuclease, they found that the Cas9 was able to successfully cleave their desired DNA target. The long-awaited, heavily-anticipated, programmable nuclease had finally arrived, and it was the product of decades of basic research. If not for the curious biologists, geneticists, and biochemists tinkering with

conserved bacterial DNA motifs and invisible-to-the-naked-eye nuclease enzymes, the revolutionary CRISPR-Cas9 technology would never have been discovered. Basic research is more than a venture to appease curious scientists; it is practically essential for future innovation and the advancement of humankind.

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